Introduction to Data Science

Class-2

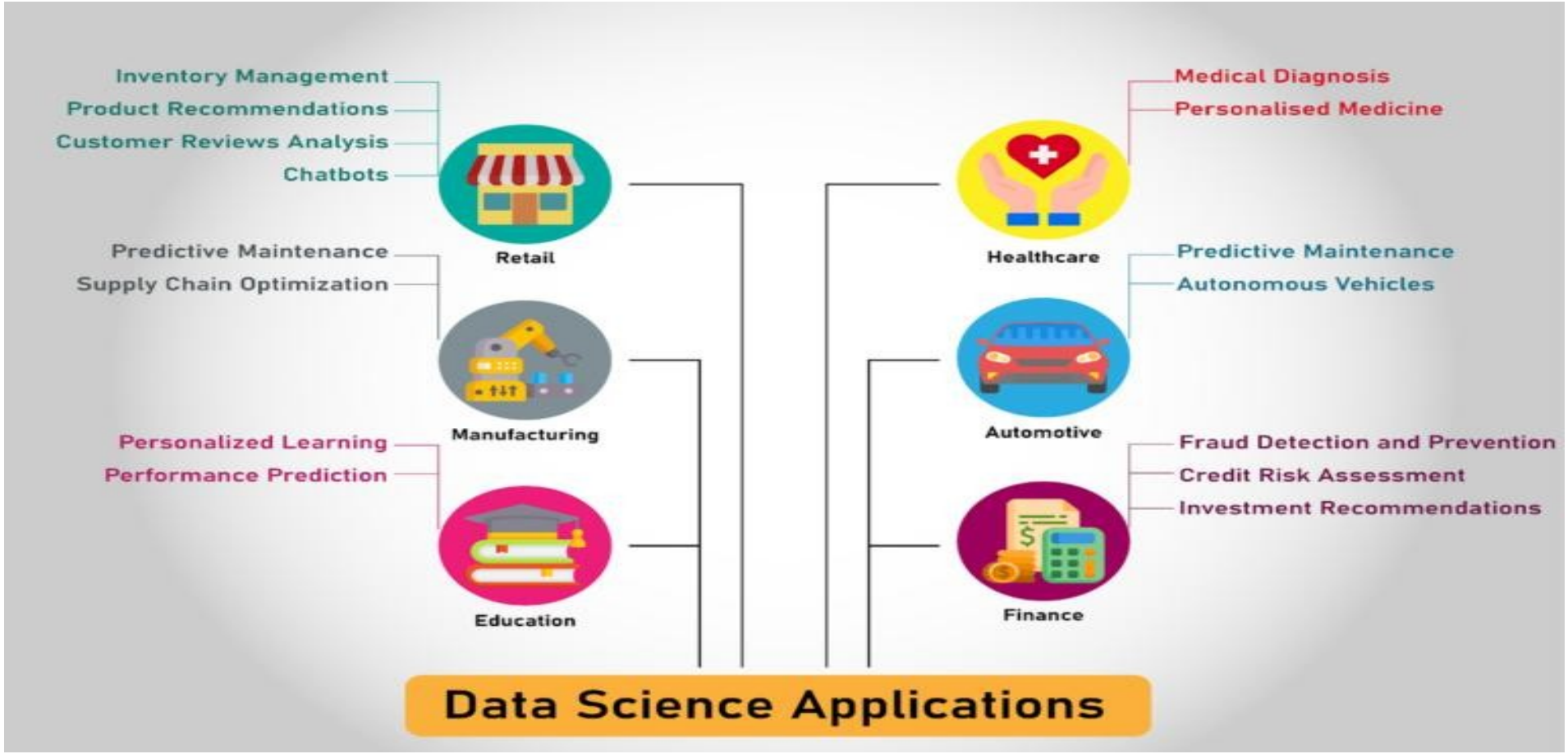
Role of a Data Scientist

- A Data Scientist transforms raw data into useful knowledge and actionable insights.
- **Example**
- A hospital has thousands of patient records.
- A Data Scientist can use those records to answer:
- Which patients are at high risk?
- What factors are associated with complications?
- How long might a patient stay in the ICU?
- Can we predict deterioration early?

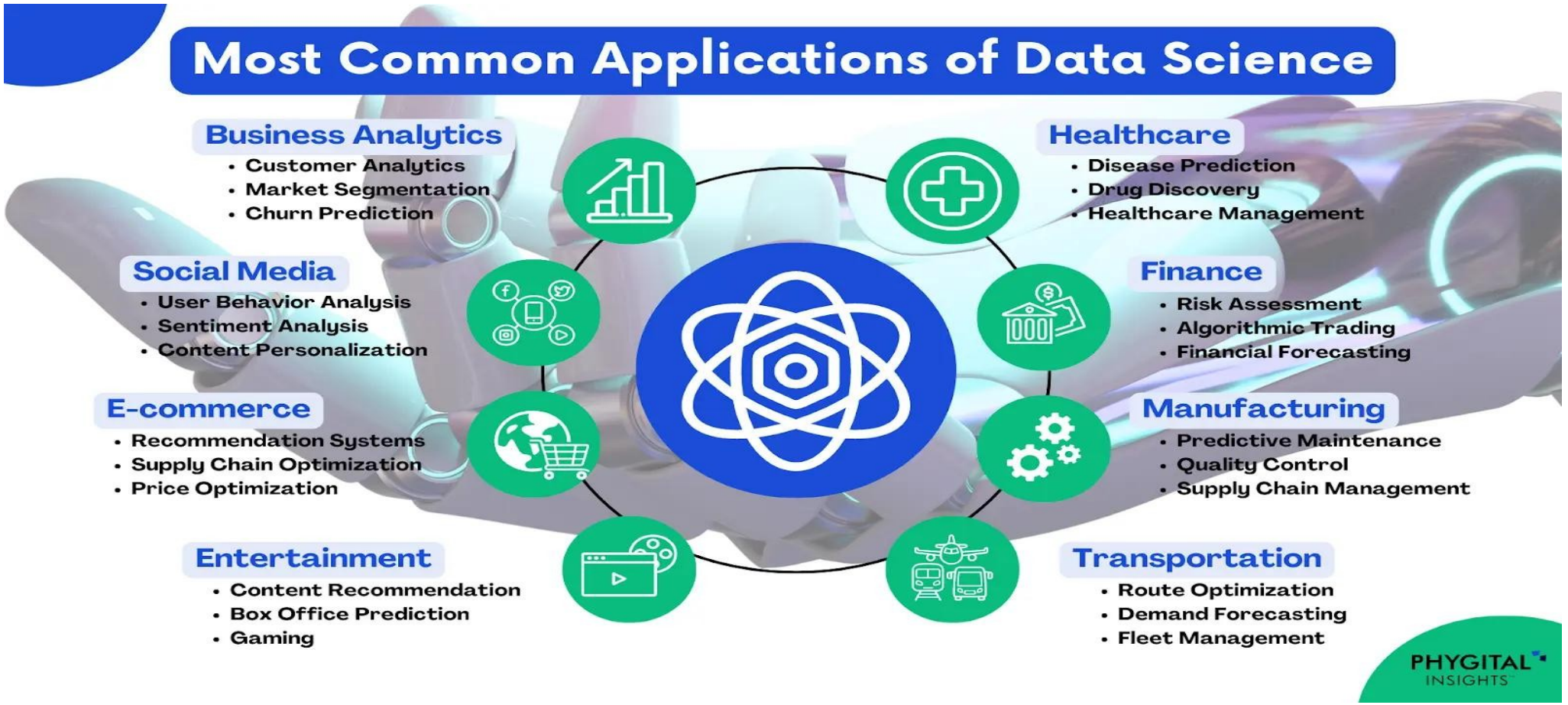
Important Skills of a Data Scientist

- **Technical Skills**
- Python/R
- SQL
- Statistics
- **Soft Skills**
- Communication
- Problem solving
- Critical thinking
- Domain knowledge
- ***A good Data Scientist is not only a programmer.They must understand the problem, data, methods, and real-world context.***

Real-World Applications of Data Science



Most Common Applications of Data Science



Business Analytics
- Customer Analytics
- Market Segmentation
- Churn Prediction

Healthcare
- Disease Prediction
- Drug Discovery
- Healthcare Management

Social Media
- User Behavior Analysis
- Sentiment Analysis
- Content Personalization

Finance
- Risk Assessment
- Algorithmic Trading
- Financial Forecasting

E-commerce
- Recommendation Systems
- Supply Chain Optimization
- Price Optimization

Manufacturing
- Predictive Maintenance
- Quality Control
- Supply Chain Management

Entertainment
- Content Recommendation
- Box Office Prediction
- Gaming

Transportation
- Route Optimization
- Demand Forecasting
- Fleet Management

PHYGITAL INSIGHTS

How is Data Science being used Today?



HealthCare

It is being used to develop new drugs and treatments, predict disease outbreaks, and improve the quality of patient care.



Finance

In Finance, it is being used to detect fraud, manage risk, and make better investment decisions.



Retail

In Retail. it is being used to understand customer behavior, predict demand, and personalize the shopping experience.



Manufacturing

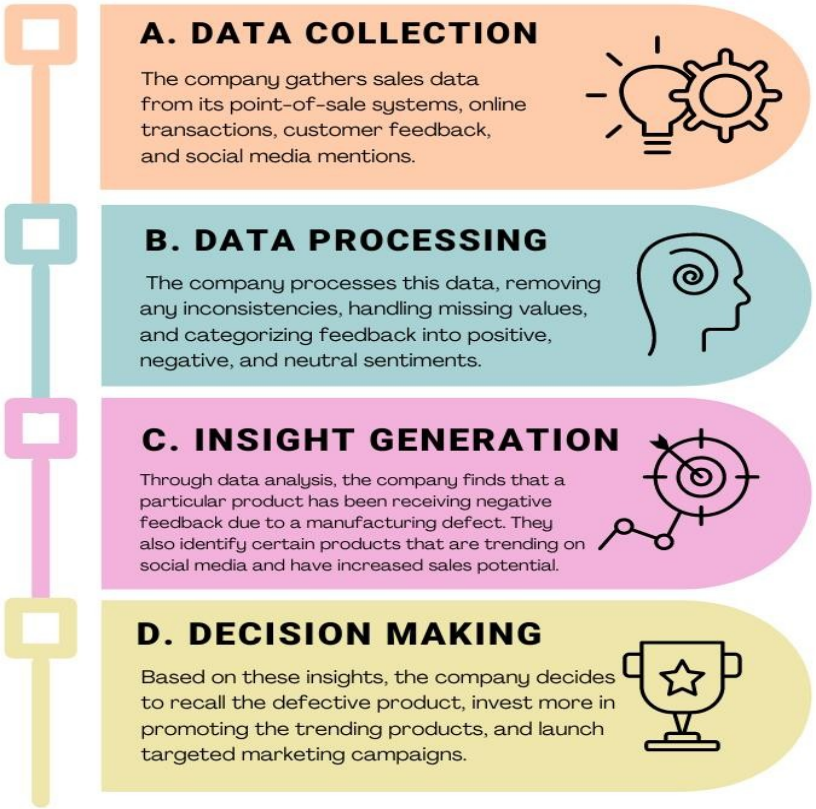
In Manufacturing, it is being used to improve product quality, reduce costs, and optimize production processes.

Data-Driven Decision Making

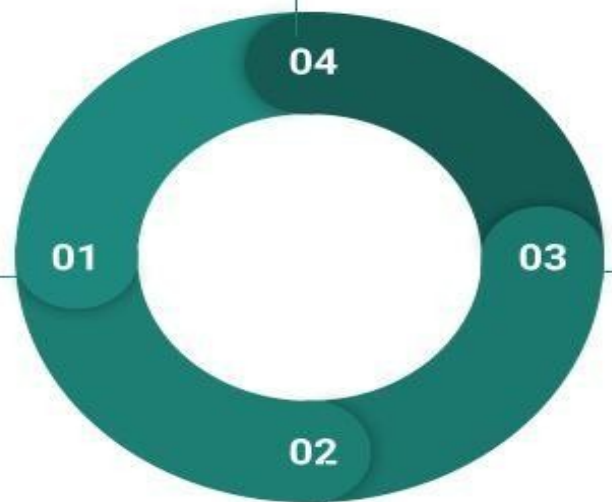
- Data-driven decision making is the process of using data and analysis to support better-informed decisions.
- ***Traditional approach***
- Experience → Opinion → Decision
- Example:"I think sales will increase next month.“
- ***Data-driven approach***
- Data → Analysis → Evidence → Decision
- Example:"Sales data from the last three years show a consistent increase during this period.“
- The second approach uses evidence from data.

Data-Driven Decision-Making Process





Decision Making
Based on analysis, make decision



Data Collection
Using Google sheet or use existing TML sheet

Data Analytics
Visualize the data in multiple way as per need

Data Cleaning
Fill the missed data or prepare the file for dashboard


Data
Insights
Actions
Outcomes
Data
+
Analytics
Decision Logic
Optimization
=
Best Decision
x
Scale
Execution
=
Results
Prediction ⟶ Prescription ⟶ Action
Adapt

Why Data-Driven Decision Making is Important

- It can help organizations:

1. Make evidence-based decisions
2. Identify patterns
3. Predict future events
4. Reduce uncertainty
5. Improve efficiency
6. Identify risks
7. Understand customers/users
8. Optimize resources

Role of a Data Scientist in Decision Making — Retail Example

- ***Scenario: An online shopping company wants to increase sales***
- The company has thousands of customers and wants to decide:
- Which products should we recommend to each customer?
- **Step 1: Collect Data**
- The Data Scientist collects:
- Customer age
- Previous purchases
- Products viewed
- Search history
- Product ratings
- Purchase frequency
- Time spent on products

- ***Step 2: Analyze Data***
- The Data Scientist looks for patterns.
- For example:Customers who purchase a laptop often purchase a laptop bag and wireless mouse.
- **Step 3: Build a Model**
- A machine-learning model predicts which products a customer is likely to purchase.
- For example:
- **Customer → Laptop purchased**
- Model predicts:
- Laptop bag → 85%
- Wireless mouse → 72%
- Keyboard → 61%

- ***Step 4: Generate Insight***
- The Data Scientist provides the insight:

"Customers who purchase laptops have a high probability of purchasing laptop accessories.“

- ***Step 5: Make a Decision***
- The company can use this information to:
- Recommend related products
- Create personalized offers
- Manage inventory
- Design marketing campaigns

Step 6: Evaluate the Decision

- After implementing recommendations, the company checks:
- Did sales increase?
- Did customers click the recommendations?
- Did the purchase rate increase?

Simple Flow

- **Customer Data**
  ↓
  Data Scientist
  ↓
  Data Analysis & Machine Learning
  ↓
  Prediction / Insight
  ↓
  Business Decision
  ↓
  Evaluate Result

Data Science and Data Mining

Aspect	**Data Science**	**Data Mining**
Definition	A broad field that uses statistics, programming, ML, and domain knowledge to extract knowledge and insights from data.	The process of discovering useful patterns, relationships, and knowledge from large datasets.
Scope	**Broad**	**Narrower**
Main goal	Understand data, make predictions, support decisions, and solve real-world problems.	Find hidden patterns and relationships in data.
Activities	Data collection → cleaning → analysis → visualization → modeling → deployment	Pattern discovery → clustering → classification → association → anomaly detection
Techniques	Statistics, ML, deep learning, data visualization, NLP, databases, etc.	Classification, clustering, association rules, anomaly detection, etc.
Output	Insights, predictions, models, dashboards, decision-support systems	Patterns, rules, groups, relationships, anomalies
Example	Build a system that predicts whether an ICU patient will deteriorate.	Discover that certain combinations of ICU variables are associated with patient deterioration.

- ***Simple example***
- Suppose a hospital has 100,000 patient records.
- Data Mining:"Can we discover patterns in these records?“
- Data Science:"How can we use these data to solve the clinical problem?“

- **Data Mining = Discover patterns from data**
- **Data Science = Use the whole data workflow to extract knowledge and solve problems**

Real-World Example

- Imagine an online shopping company such as Amazon.
- ***Data Mining***
- The company analyzes customer transactions and discovers:
- Customers who buy a laptop often buy a laptop bag.This is pattern discovery.
- ***Data Science***
- The company can use that pattern to develop a recommendation system:"Customers who bought this laptop may also be interested in this laptop bag.“
- It may also collect data, build a recommendation model, evaluate it, deploy it, and monitor its performance.

Data Warehouse

- A **Data Warehouse (DW)** is a centralized system that stores **large amounts of historical data collected from different sources**. It is mainly used for **analysis, reporting, and decision-making**, rather than for day-to-day transactions.
- **Simple Example: University**
- University has data stored in different systems:
- 🎓Student Information System → student records
- 📚Library System → book borrowing
- 💰Finance System → tuition payments
- 📝Examination System → grades

- A **Data Warehouse** can bring the data together:
- **Student System** ─┐
  Library System ─┤
  Finance System ─┼─→ **DATA WAREHOUSE** → Analysis & Reports
  Exam System ─┘
- Now the university can ask:
- "What is the relationship between students' attendance, library usage, and academic performance?"

Data Warehouse and Data Science

- A Data Warehouse can provide **organized, historical data** for Data Scientists.

Data Sources
↓
ETL / Data Integration
↓
Data Warehouse
↓
Data Analysis / Data Mining
↓
Machine Learning / Data Science
↓
Insights & Decisions